Summary in Graph

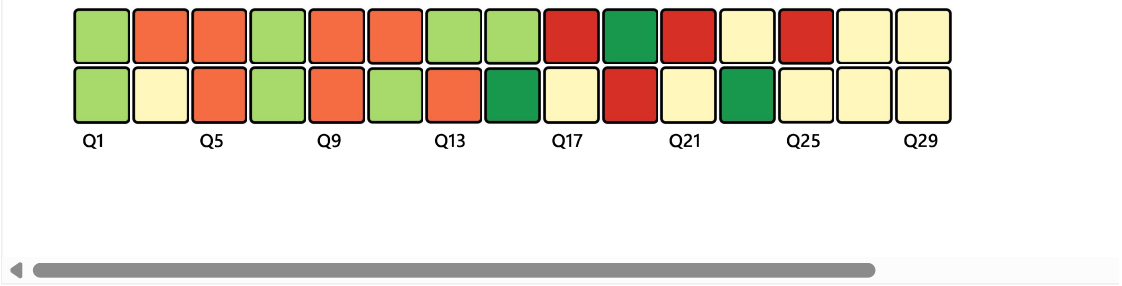
Exam Summary (GO Classes DA Test Series 2026 | Linear Algebra | Subject Wise Test 1)

Qs. Attempted:	25 (15 + 10)	Correct Marks:	13 (7 + 6)
Correct Attempts:	10 (7 + 3)	Penalty Marks:	5 (2.33 + 2.67)
Incorrect Attempts:	15 (8 + 7)	Resultant Marks:	8 (4.67 + 3.33)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	105 Minutes
Time Taken:	105 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (36) out of 359 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ real matrix.
Consider the following statements regarding its fundamental subspaces and mark all the statement(s) that is/are TRUE:

A. If W is the column space of A, then $W^{\perp}$ is the null space of A^T.
B. The null space of A is a subspace of $\mathbb{R}^n$.
C. The null space of A is a subspace of $\mathbb{R}^m$.
D. The column space of A is a subspace of $\mathbb{R}^m$.

Your Answer: A;B;D | Correct Answer: A;B;D | Correct | Time taken: 03min 16sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

Which of the following are the singular values of matrix A ?

A. $\sqrt{5}, \sqrt{5}$
B. $3,1$
C. $2\sqrt{2}, \sqrt{2}$
D. $4,2$

Your Answer: C | Correct Answer: C | Correct | Time taken: 03min 14sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let B be a square matrix such that $C = \text{adj}(B)$, where $\text{adj}(B)$ denotes the adjugate of B. Suppose $|B| \neq 0$. What is the value of the determinant of the matrix $C^{-1}B^2$?

A. $|B|$

B. $|B|^2$

C. $\frac{1}{|B|^{n-3}}$

D. $\frac{1}{|B|^2}$

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 04min 06sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

A 10×13 matrix A is given and the homogeneous system $A\vec{x} = \overrightarrow{0}$ is transformed to reduced row echelon form. There are 7 lead variables.
How many free variables?

Your Answer: 7 | Correct Answer: 6 | Incorrect | Time taken: 02min 26sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

A system $A\mathbf{x} = \mathbf{b}$ with $m \times 2024$ matrix A and $\mathbf{b} = (1, 0, \ldots, 0)^T$ has a unique solution. Which of the these statements does not follow from these conditions?

A. $\text{rank}(A) = 2024$
B. $m \geq 2024$
C. Columns of A are linearly independent
D. A is invertible

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 02min 41sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

What is the projection of the vector $(1, 2)^T$ on the line $x = y$ in $\mathbf{R}^2$?

A. $(3/2, 3/2)^T$
B. $(2, 2)^T$
C. $(2, 1)^T$
D. None of the above

Your Answer: B | Correct Answer: A | Incorrect | Time taken: 02min 52sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a 4×4 matrix in which the first two rows are not proportional (none of them is a multiple of another), the third row is the sum of the first two, and the fourth row is the difference of the first two. What are the dimensions of the four subspaces $R(A), C(A), N(A)$ and $N\left(A^T\right)$ (in this order)?

A. $3, 3, 0, 4$
B. $2, 2, 2, 2$
C. $3, 3, 0, 0$
D. $1, 1, 3, 2$

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 57sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a 4×4 matrix. If $\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$ are solutions of the system of linear equations $A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$, then which of the following is also a solution?

A. $\mathbf{p} - \mathbf{q}$

B. $\mathbf{q} - 2\mathbf{p}$

C. $2\mathbf{p} - \mathbf{q}$

D. All of the above

Your Answer: C | Correct Answer: C | Correct | Time taken: 02min 58sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Assume that the 4×4 matrix A is row equivalent to the matrix B, where

$$B = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Which of the following statements is true?

A. B is not the reduced row echelon form of A.
B. $\det(A) \neq 0$.
C. The null space of A has dimension 3 .
D. If $A\vec{x} = \vec{b}$ is consistent for some $\vec{b} \neq \overrightarrow{0}$, then it has infinitely many solutions.

Your Answer: D | Correct Answer: A | Incorrect | Time taken: 06min 14sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Which of the following statements about all 5×5 matrices is true?

A. $\det(A + B) = \det A + \det B$
B. $\det(-A) = -\det(A)$.
C. $AB = 0$ implies $A = 0$ or $B = 0$.
D. If $\det A = 0$ then some two rows are proportional.

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 02min 14sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = \begin{pmatrix} 2 & -1 & -2 \\ -4 & 2 & 4 \\ -8 & 4 & 8 \end{pmatrix}$. Then the following is a basis of the nullspace of A.

A. $\left\{ \begin{bmatrix} 2 & -4 & -8 \end{bmatrix}^T, \begin{bmatrix} -1 & 2 & 4 \end{bmatrix}^T, \begin{bmatrix} -2 & 4 & 8 \end{bmatrix}^T \right\}$

B. $\left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T, \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}^T \right\}$

C. $\left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T, \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}^T, \begin{bmatrix} 3 & 2 & 2 \end{bmatrix}^T \right\}$

D. $\left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \right\}$

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 09min 31sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let

$$V = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_1 + x_2 + x_3 + x_4 = 0 \right\}$$

be a subspace of $\mathbb{R}^4$. What is the dimension of the subspace V ?

Your Answer: 3 | Correct Answer: 3 | Correct | Time taken: 01min 56sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Which of the following could be a non-zero singular value for matrix B below?

$$B = \begin{bmatrix} 1 & 5 & 1 & 1 & 2 \\ 2 & 7 & 2 & 9 & 4 \\ 3 & 3 & 3 & 4 & 6 \end{bmatrix}$$

A. $1.01 + 2.14\text{j}$
B. -1.05
C. -100
D. 4.04

Your Answer: D | Correct Answer: D | Correct | Time taken: 01min 22sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the following two statements:
$S1$: If a matrix has determinant zero, then two of its columns are scalar multiples of each other, or one of the columns is the zero vector.

$S2$: If λ is an eigenvalue of $AA^\top$, then λ is necessarily an eigenvalue of $A^\top A$.
Which of the following is correct?

A. $S1$ is true, $S2$ is false

B. $S1$ is false, $S2$ is true

C. Both $S1$ and $S2$ are true

D. Both $S1$ and $S2$ are false

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 00min 41sec | Discuss

Q #15 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following matrices defined using real numbers a, b, c :

$$A = \begin{bmatrix} 1 & a \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & b \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & c \\ 1 & 1 \end{bmatrix}$$

Each matrix is of the form $\begin{bmatrix} 1 & x \\ 1 & 1 \end{bmatrix}$, where $x \in \mathbb{R}$.
Which of the following statements is/are TRUE?

A. Matrix A has two distinct real eigenvalues if and only if $a > 0$.
B. Matrix B has a repeated real eigenvalue if and only if $b = 0$.
C. Matrix C has complex (non-real) eigenvalues if and only if $c < 0$.

D. The eigenvalues of all three matrices are real for any real values of a, b, c.

Your Answer: A;B;C | Correct Answer: A;B;C | Correct | Time taken: 03min 57sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by:

$$T\left(\begin{bmatrix}2\\1\end{bmatrix}\right) = \begin{bmatrix}4\\3\end{bmatrix}, \quad T\left(\begin{bmatrix}1\\3\end{bmatrix}\right) = \begin{bmatrix}4\\1\end{bmatrix}$$

Which of the following matrices A satisfies $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^2$?

A. $\frac{1}{5}\begin{bmatrix}5 & 0\\0 & 5\end{bmatrix}$

B. $\frac{1}{5}\begin{bmatrix}8 & 4\\4 & -1\end{bmatrix}$

C. $\frac{1}{5}\begin{bmatrix}8 & 4\\8 & -1\end{bmatrix}$

D. $\begin{bmatrix}4 & 4\\1 & 3\end{bmatrix}$

Your Answer: C | Correct Answer: C | Correct | Time taken: 07min 35sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let V be a vector space, and let W_1 and W_2 be subspaces of V. Define the set

$$U = \{\mathbf{v} \in V : \mathbf{v} \in W_1 \text{ and } \mathbf{v} \in W_2\}$$

Which of the following statements is true about the set U ?

A. U is not a subspace of V unless $W_1 = W_2$.
B. U is a subspace of V because the intersection of two subspaces is always a subspace.
C. U is a subspace of V only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
D. U may not contain the zero vector, so it is not always a subspace.

Your Answer: D | Correct Answer: B | Incorrect | Time taken: 10min 51sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ matrix. Let S_1 be the row space of A, and S_2 be the column space of A. Define a transformation

$$T : S_1 \to S_2 \quad \text{by} \quad T(\vec{x}) = A\vec{x}$$

Which of the following statements is/are TRUE?

A. T is one-to-one because if $A\vec{x}_1 = A\vec{x}_2$, then $\vec{x}_1 = \vec{x}_2$ for all $\vec{x}_1, \vec{x}_2 \in S_1$.
B. T is onto because every vector in the column space of A can be written as $A\vec{x}$ for some $\vec{x} \in S_1$.
C. T is not one-to-one because row vectors might map to the same column vector.
D. T is a bijection between the row space and the column space of A.

Your Answer: A;C | Correct Answer: A;B;D | Incorrect | Time taken: 11min 28sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $A = LU$, where L is lower triangular with 1's on the main diagonal, and $U = \left(u_{i,j}\right)$ is upper triangular, and

$$A = \begin{pmatrix} 2 & 1 & 3 & 0 \\ -4 & 0 & -5 & 0 \\ 2 & 5 & 5 & 2 \end{pmatrix}$$

What's $u_{2,4}$?

A. 2
B. -1
C. 1
D. 0

Your Answer: D | Correct Answer: D | Correct | Time taken: 02min 37sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let the following vectors in $\mathbb{R}^2$ be given:

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathcal{C} = \{\mathbf{u}_1, \mathbf{u}_2\}$ be ordered bases for $\mathbb{R}^2$.
What is the change-of-basis matrix that converts coordinates from the basis $\mathcal{B}$ to the basis $\mathcal{C}$?

A. $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 10 & 11 \\ 4 & 4 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 03min 10sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

A is an $m \times n$ matrix and $\mathbf{b}$ is an $m \times 1$ vector. The equation $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. Consider the following statements:
(i) $m \leq n$

(ii) $n \leq m$

(iii) the rank of $A = n$

(iv) the rank of $A < n$

(v) $\det A = 0$

Which must be true?

A. only (i) and (v)
B. only (iv)
C. only (v)
D. only (iii) and (v)

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 03min 42sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following statements related to properties of matrices and subspaces in $\mathbb{R}^n$.
Which of the following statements is/are TRUE?

A. If the rows of a matrix $A \in \mathbb{R}^{m \times n}$ are linearly independent, then for every $\vec{b} \in \mathbb{R}^m$, the system $A\vec{x} = \vec{b}$ has at most one solution $\vec{x} \in \mathbb{R}^n$.
B. If the columns of a square matrix A are linearly independent, then the transpose A^T is invertible.
C. If V and W are subspaces of $\mathbb{R}^n$, then the union $V \cup W$ is also a subspace of $\mathbb{R}^n$.
D. If A and B are square matrices and AB is invertible, then both A and B are invertible.

Your Answer: A;B;C;D | Correct Answer: B;D | Incorrect | Time taken: 04min 47sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{n \times n}$ be a real square matrix with singular values $\sigma_1, \sigma_2, \ldots, \sigma_n$.
Which of the following statements is/are TRUE?

A. The singular values of A^2 are necessarily equal to $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$.
B. The singular values of $A^T A$ are necessarily equal to $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$.
C. The singular values of $A^T A$ are the eigenvalues of $A^T A$, which are always non-negative.
D. The eigenvalues of A and the singular values of A are always equal.

Your Answer: A;B;C | Correct Answer: B;C | Incorrect | Time taken: 04min 21sec | Discuss

Q #24 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements about the Singular Value Decomposition (SVD) are true when written in the form $A = \sigma_1 \overrightarrow{u_1}\overrightarrow{v_1}^\top + \sigma_2 \overrightarrow{u_2}^\top \overrightarrow{v_2}^\top + \cdots$? Assume that all σ_i, the singular values, are non-zero.

A. $\left\{\overrightarrow{u_1}, \overrightarrow{u_2}, \ldots\right\}$ is an orthonormal basis for the column space of $\boldsymbol{A}$.

B. The singular values, σ_i, are real numbers of arbitrary sign.

C. The SVD separates a rank $\boldsymbol{r}$ matrix $\boldsymbol{A}$ into a sum of $\boldsymbol{r}-1$ rank 1 matrices.

D. The SVD of a matrix $\boldsymbol{A}$ is unique.

Your Answer: A | Correct Answer: A | Correct | Time taken: 03min 12sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following two statements:

- **S1** : If $\{\mathbf{q}_1, \mathbf{q}_2\}$ is an orthonormal set, then the matrix $\mathbf{q}_1\mathbf{q}_1^\top + \mathbf{q}_2\mathbf{q}_2^\top$ is a projection matrix.
- **S2**: If $\mathbf{A}$ is symmetric, then every vector $\mathbf{x} \in \mathbb{R}^n$ can be uniquely written as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$, where $\mathbf{x}_1 \in \mathcal{N}(\mathbf{A}), \mathbf{x}_2 \in \mathcal{C}(\mathbf{A})$.

Which of the following is correct?

A. $S1$ is true, $S2$ is false

B. $S1$ is false, $S2$ is true

C. Both $S1$ and $S2$ are true

D. Both $S1$ and $S2$ are false

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 03min 37sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Find the decomposition

$$\mathbf{v} = \mathbf{v}_{\text{null}} + \mathbf{v}_{\text{row}}$$

where $\mathbf{v}_{\text{null}} \in \mathcal{N}(A)$ (null space of A) and $\mathbf{v}_{\text{row}} \in \mathcal{R}(A)$ (row space of A).

A. $\mathbf{v}_{\text{null}} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_{\text{row}} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$

B. $\mathbf{v}_{\text{null}} = \frac{1}{3}\begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_{\text{row}} = \frac{1}{3}\begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$

C. $\mathbf{v}_{\text{null}} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_{\text{row}} = \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix}$

D. $\mathbf{v}_{\text{null}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_{\text{row}} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $A = \begin{bmatrix} 1 & a \\ 1 & 1 \end{bmatrix}$, where $a > 0$.

Suppose A is diagonalizable as $A = VDV^{-1}$, where D is a diagonal matrix of eigenvalues and V is the matrix of corresponding eigenvectors.

Which of the following is a valid choice for the matrix V ?

A. $V = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

B. $V = \begin{bmatrix} \sqrt{a} & -\sqrt{a} \\ 1 & 1 \end{bmatrix}$

C. $V = \begin{bmatrix} 1 & 1 \\ \sqrt{a} & -\sqrt{a} \end{bmatrix}$

D. $V = \begin{bmatrix} a & -a \\ 1 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The matrix A has the following Singular Value Decomposition (SVD):

$$A = \underbrace{\frac{1}{5}\begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}}_{U} \cdot \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\Sigma} \cdot \underbrace{\frac{1}{3}\begin{bmatrix} 1 & x & 2 \\ 2 & 2 & y \\ 2 & -1 & -2 \end{bmatrix}}_{V^T}$$

Here, U and V are square matrices, and $x, y \in \mathbb{R}$ are unknown real numbers.
Which of the following pairs of values for x and y is consistent with the decomposition?

A. $x = 2, \quad y = -1$
B. $x = -2, \quad y = 1$
C. $x = -1, \quad y = 2$
D. $x = 1, \quad y = -2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{3\times 3}$ be a symmetric matrix with distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3$ and corresponding orthonormal eigenvectors $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$. Which of the following correctly describes the Singular Value Decomposition (SVD) of A ?

A. If $\lambda_1, \lambda_2, \lambda_3 \geq 0$, then

$$\text{SVD of } A = Q\Sigma Q^T \quad \text{where } Q = [\boldsymbol{q}_1 \boldsymbol{q}_2 \boldsymbol{q}_3], \quad \Sigma = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$$

B. If $\lambda_1, \lambda_2 > 0 > \lambda_3$, then

$$\text{SVD of } A = U\Sigma V^T \quad \text{where } U = [\boldsymbol{q}_1 \boldsymbol{q}_2 - \boldsymbol{q}_3], \quad V = Q, \quad \Sigma = \text{diag}(|\lambda_1|, |\lambda_2|, |\lambda_3|)$$

C. For any symmetric matrix with eigenvalue decomposition $A = Q\Lambda Q^T$, the SVD is

$$\text{SVD of } A = Q|\Lambda|Q^T \quad \text{where } |\Lambda| = \text{diag}(|\lambda_1|, |\lambda_2|, |\lambda_3|)$$

D. If $\lambda_3 = 0$, then one of the singular values of A is 0 , and the corresponding singular vectors can be taken as $\boldsymbol{q}_3$ (for both U and V).

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -2 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Let $x \in \mathbb{R}^2$ be any vector, and consider the quantity $\|Ax - b\|_2$. What is the minimum value of this expression over all possible $x \in \mathbb{R}^2$?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

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